

HOMEWORK 3
BIOSTATISTICS 794
DUE OCTOBER 30TH, 2025

Instructions

- Submit a single .Rmd or .html/.pdf file with code, figures, and interpretations.
- Clearly label each section and question.
- Use cross-validation wherever possible, with appropriate metrics.
- Comment your R code and explain your interpretations in plain language.

Learning Goals By the end of this assignment, you should be able to:

- Apply and interpret Naive Bayes classifiers with mixed data types.
 - Grow and prune CART models for regression and classification.
 - Use ensemble methods (Random Forests) and interpret variable importance and partial dependence.
 - Fit a Support Vector Machine (SVM) and evaluate its predictive ability.
 - Fit and evaluate a small feed-forward neural network.
1. For this question, use the **Heart Disease** dataset (`mlbench::Heart`) or an equivalent from the UCI repository.
 - (a) Explore the dataset and describe the variables (which are continuous, categorical, or binary?).
 - (b) Split the data into training (80%) and testing (20%) sets.
 - (c) Fit a **Naive Bayes** classifier using all predictors. Use `train(..., method="naive.bayes")` from the `caret` package.
 - (d) Evaluate AUC, accuracy, sensitivity, specificity, PPV, and NPV on the test set.
 - (e) Repeat the analysis after converting some variables to categorical (e.g., bin cholesterol or age). Compare performance.
 - (f) Briefly discuss when Naive Bayes is advantageous in public health prediction.
 2. Use the **birthwt** dataset from the `MASS` package.
 - (a) Describe the data and the outcome variable (`bwt`).
 - (b) Fit a **CART regression tree** using `rpart(bwt ~ ., data=birthwt)`.
 - (c) Use cross-validation to select the best complexity parameter (`cp`).

- (d) Plot the tree and interpret the top two splits in the context of maternal health.
 - (e) Compare model RMSE to a standard multiple linear regression model.
 - (f) (Optional) Discuss how overfitting might appear in a regression tree.
3. Use the **NHANES** dataset (via `nhanesA` package) or **Air Quality** (`datasets::airquality`) if NHANES is unavailable. Choose a continuous outcome such as BMI or Ozone.
- (a) Split the data into training (70%) and test (30%) datasets. Preprocess the data: handle missing values, and normalize predictors using `recipes`.
 - (b) Fit a **Random Forest** regression model using `randomForest` (or `ranger`) for a fixed `ntree` and `mtry`.
 - (c) Tune `mtry` using cross-validation.
 - (d) Plot variable importance. Identify the top three predictors and interpret them.
 - (e) Create and interpret a **partial dependence plot (PDP)** for one important variable.
 - (f) Use the Radial Basis Function (RBF) kernel to fit an SVM to the data. Tune the cost using CV (you can use the default value of sigma).
 - (g) Compare the Random Forest and SVM test RMSE.
4. Use the **BreastCancer** dataset from `mlbench` (binary outcome: benign vs. malignant).
- (a) Preprocess the data: handle missing values, and normalize predictors using `recipes`.
 - (b) Fit a single hidden-layer neural network using the `nnet` or `keras` package.
 - (c) Tune the number of hidden units (e.g., 2, 5, 10) and weight decay (regularization).
 - (d) Compare performance (AUC, accuracy) across models using 5-fold cross-validation.
 - (e) Evaluate the best model on the test set and compare it to a logistic regression and a random forest.
 - (f) Discuss: When might neural networks be unnecessary or overkill for public health data?

Submission Checklist

- ☐ Code runs cleanly and results are reproducible.
- ☐ Each figure is clearly labeled and interpreted.
- ☐ Training/test or CV procedures are correctly described.
- ☐ Discussion focuses on interpretation, not just numeric results.